



RENGGO

Live Escalation War Room

Mid-call multi-agent collaboration with Band SDK for complex customer support

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Built for the Band of Agents Hackathon

THE PROBLEM

Frontline bots win the routine — and hit a wall on the hard calls



The Routine AI Success

Frontline voice systems like Renggo excel at low-complexity, real-time tasks.

- Answering FAQs
- Authenticating users
- Retrieving accounts

They fail because they are single-threaded and lack context reasoning.



The High-Stakes Wall



Policy-Heavy Queries

Resolving disputes requires searching vast, conflicting company guidelines.



Compliance Risks

Uncontrolled LLMs risk violating legal regulations or promising unauthorized compensation.



Human Authorization Needed

High-value actions need human approval, but voice bots cannot wait on a phone call.

THE SOLUTION

Safe & Dynamic Delegation

When a caller has a complex problem, Renggo delegates the query to a private backend “War Room” using the Band SDK.

The live bot keeps the customer engaged with natural callback updates, while specialized agents debate, check compliance, and seek authorization — asynchronously.

The War Room

Four specialized entities, one resolution.



Triage Agent

LangGraph

Classifies the escalation, flags severity, and outputs a `requires_human_approval` flag.



Knowledge Agent

RAG Retrieval

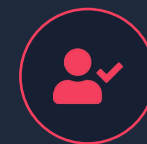
Audits policy manuals, finds matching guidelines, and assigns a confidence score.



Compliance Agent

GPT-4o-mini

Double-checks proposed solutions against corporate rules to ensure safe execution.

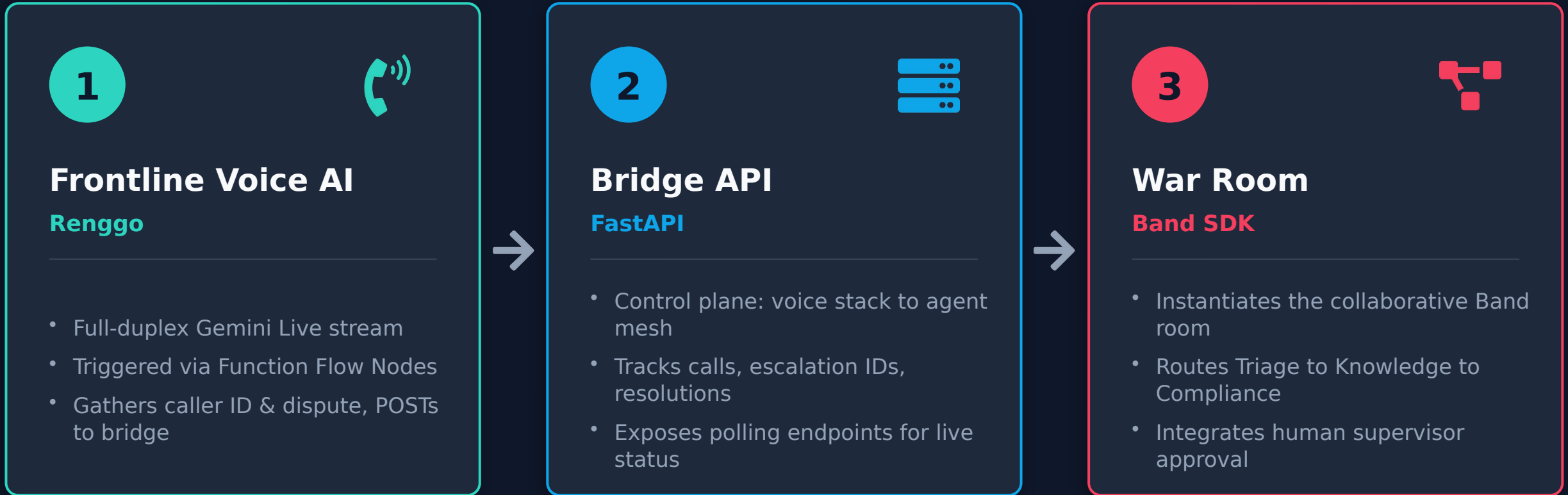


Supervisor UI

Human-in-the-Loop

A web portal letting human managers authorize high-value resolutions in one click.

From voice channel to agentic mesh



Flow: Renggo triggers the Bridge API via a Function Node → Bridge initiates the Band room → Agents resolve → Renggo polls and reads the resolution.

Test it live — two escalation paths



TEST IT LIVE

Call

+1 (808) 720-2508

Dial in to experience the full-duplex integration. Talk to a live voice agent, state a dispute, and witness the backend orchestration kick in.



PATH 1 — THE FAST PATH

Billing Dispute

"I was double charged in March."

Triage classifies as `billing_dispute` (`severity=low`, `requires_approval=false`). Knowledge RAG resolves immediately. Compliance validates. Bot speaks the resolution in-call (~15s).



PATH 2 — THE SLOW PATH

Hardship Refund

"I lost my job and cannot pay this month's bill."

Triage classifies as `hardship_refund` (`severity=high`, `requires_approval=true`). Voice agent promises a callback and clears the line. Backend continues asynchronously. Supervisor approves via Band UI.

Where this goes next



Generic Architecture

Infinite Domains

The bridge and multi-agent setup are entirely domain-agnostic. Swap the knowledge base and set new compliance constraints to instantly power insurance, travel, or medical support.



Safe LLM Execution

Guarded Outcomes

A dedicated compliance agent alongside human-in-the-loop approvals removes the business risk of rogue LLM actions — giving enterprise operators total trust.



Future of Voice Support

True Automation

Unifying low-latency full-duplex voice with deep asynchronous multi-agent reasoning creates the first support setup capable of complex logic in real-time.